02/09/19

Cayley Hamilton Theorem

To verify cayley hamilton theorem find characteristic eqⁿ.

$$|A - \lambda I| = 0$$

$$(-1)^n \lambda^n + K_1 \lambda^{n-1} + K_2 \lambda^{n-2} + \dots + K_n = 0$$

Replace 'λ' by 'A'

$$\boxed{(-1)^n A^n + K_1 A^{n-1} + K_2 A^{n-2} + \dots + K_n I = 0}$$

Application of cayley Hamilton theorem

It is used to solve higher degree polynomials of A and to find inverse.

Q. Verify Cayley hamilton theorem and also find its inverse.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Sol. Characteristic eqⁿ $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 3 \\ 2 & 1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{vmatrix} = 0$$

$$\boxed{(-1)^3 \lambda^3 + 3\lambda^2 + \lambda - 9 = 0} \rightarrow \text{Characteristic eqⁿ}$$

Replace by A

$$\boxed{-A^3 + 3A^2 + A - 9I = 0} \text{ Cayley hamilton theorem.}$$

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or $A^3 - 3A^2 - A + 9I = 0$

Solving LHS

$$A^2 = A \cdot A = \begin{bmatrix} 4 & -3 & 6 \\ 3 & -2 & 5 \\ 0 & -2 & 5 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 1 & -7 & 7 \end{bmatrix}$$

Now

$$A^3 - 3A^2 - A + 9I$$

$$= \begin{bmatrix} 4 & -9 & 21 \\ 11 & -2 & 11 \\ 1 & -7 & 7 \end{bmatrix} - \begin{bmatrix} 12 & -9 & 18 \\ 9 & 6 & 12 \\ 0 & -6 & 15 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 0$$

Hence verified (Cayley Hamilton theorem)

Now finding out inverse
Pre multiplying by A^{-1}

$$A^2 - 3A - I + 9A^{-1} = 0$$

$$9A^{-1} = -A^2 + 3A + I$$

$$A^{-1} = \frac{1}{9}\left[-A^2 + 3A + I\right]$$

$$A^{-1} = \frac{1}{9}\left\{\begin{bmatrix} -4 & 3 & -4 \\ -3 & -2 & -4 \\ 0 & 2 & -5 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 9 \\ 6 & 3 & -3 \\ 3 & -3 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right\}$$

$$A^{-1} = \frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -7 \\ 3 & -1 & -1 \end{bmatrix}$$

Q. If $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ express

$$A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$$

as linear polynomial.

Sol. Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) + 2 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 + 2 = 0$$

$$\lambda^2 - 4\lambda + 5 = 0$$

Replace λ by A

$$\boxed{A^2 - 4A + 5I = 0}$$

$$A^4 + 3A^2 - I$$

$$A^2 - 4A + 5I \overline{\big) A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2}$$

$$A^6 - 4A^5 + 5A^4$$

$$3A^4 - 12A^3 + 14A^2$$

$$3A^4 - 12A^3 + 15A^2$$

$$-A^2$$

$$-A^2 + 4A - 5I$$

$$(-) \quad (+)$$

$$-4A + 5I$$

$$A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2 = (A^2 - 4A + 5I)(A^4 + 3A^2 - I) + 5I - 4A$$

$$= 0 + 5I - 4A \quad \text{(Linear polynomial)}$$

$$= 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 4\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -8 \\ 4 & -7 \end{bmatrix}$$

Q. Find the characteristic equation of matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Solve the equation $A^5 + 5A^4 - 6A^3 + 2A^2 - 4A + 7I$ and also find the inverse of this matrix.

Sol. Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$(3 - \lambda - 3\lambda + \lambda^2) - 8 = 0$$

$$\boxed{\lambda^2 - 4\lambda - 5 = 0} \rightarrow \text{Characteristic } eq^n$$

Replace λ by A

$$\boxed{A^2 - 4A - 5 = 0} \rightarrow \text{Cayley Hamilton Theorem}$$

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$$
\begin{array}{r|l}
 & A^3 + 9A^2 + 35A + 18I \\
\hline
A^2 - 4A - 5I & A^5 + 5A^4 - 6A^3 + 2A^2 - 4A + 7I \\
 & A^5 - 4A^4 - 5A^3 \\
\hline
 & 9A^4 - A^3 + 2A^2 - 4A + 7I \\
 & 9A^4 - 36A^3 - 45A^2 \\
\hline
 & 35A^3 + 44A^2 - 4A + 7I \\
 & 35A^3 - 140A^2 - 175A \\
\hline
 & 187A^2 + 171A + 7I \\
 & 187A^2 - 748A - 935I \\
\hline
 & 919A + 942I
\end{array}
$$

$A^5 + 5A^4 - 6A^3 + 2A^2 - 4A + 7I$

$= (A^2 - 4A - 5I)(A^3 + 9A^2 + 35A + 187) + 919A + 942I$

$= 919A + 942I$

$= 919\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} + 942\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 919 & 3676 \\ 1838 & 2757 \end{bmatrix} + \begin{bmatrix} 942 & 0 \\ 0 & 942 \end{bmatrix}$

$= \begin{bmatrix} 1861 & 3676 \\ 1838 & 3699 \end{bmatrix}$

$A^2 - 4A - 5 = 0$

Pre multiply by A^{-1}

$A - 4I - 5A^{-1} = 0$

$-5A^{-1} = -A + 4$

$5A^{-1} = A - 4I$

$A^{-1} = \frac{1}{5}[A - 4I]$

$A^{-1} = \frac{1}{5}\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4I$

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5}\begin{bmatrix} -3 & 4 \\ 2 & -2 \end{bmatrix}$$

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Diagonalised Form of Matrix

Diagonal matrix, $D = B^{-1}AB$

where B is modal or matrix of Eigen vectors

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$B = \begin{bmatrix} X_1 & X_2 & X_3 \end{bmatrix}$$

Q. Diagonalise the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

Sol. Charaterristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$6 - 2 = 4$
$3 + 2 = 5$
2
4

$-1(2-4)$
2

$$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

$$\boxed{\lambda = 1, 2, 3}$$

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For $\lambda = 1$

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

$$0x_1 + 0y_1 - z_1 = 0$$
$$x_1 + y_1 + z_1 = 0$$
$$2x_1 + 2y_1 + 2z_1 = 0$$

$$z_1 = 0$$
$$x_1 + y_1 = 0$$
$$x_1 = -y_1$$
$$\frac{x_1}{-1} = \frac{y}{1}$$

$[(-1), (1), 0] \rightarrow$ Ist Eigen vector

For $\lambda = 2$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = 0$$

$$-x_2 + 0y_2 - z_2 = 0$$
$$x_2 + 0y_2 + z_2 = 0$$
$$2x_2 + 2y_2 + z_2 = 0$$

$$\frac{x_2}{-2} = \frac{y_2}{2-1} = \frac{z_2}{2}$$

$(-2, 1, 2) \rightarrow$ IInd Eigen vector

$$* \quad B^{-1}AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = D$$

Eigen values in diagonal

For $\lambda = 3$

$$-2x_3 + 0y_3 - z_3 = 0$$
$$x_3 - y_3 + z_3 = 0$$
$$2x_3 + 2y_3 + 0z_3 = 0$$

$$\frac{x_3}{-1} = \frac{y_3}{-1+2} = \frac{z_3}{2}$$

$(-1, 1, 2) \rightarrow$ IIIrd Eigen vector

$$B = \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow \text{Modal Matrix}$$

$\downarrow$ Ist Eigen vector $\quad \downarrow$ IInd Eigen vector $\quad \downarrow$ IIIrd Eigen vector

Power of Matrix

$$B^{-1}AB = D$$

$$(B^{-1}AB)(B^{-1}AB) = D^2$$

$$B^{-1}A^2B = D^2$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$\boxed{B^{-1}A^nB = D^n}$$

Pre multiply by B & Post multiply by B^{-1}

$$\boxed{A^n = BD^nB^{-1}} \rightarrow \text{Power of matrix}$$

$$A^4 = B\begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 81 \end{bmatrix}B^{-1} \quad \text{where}$$

Reduce the eqn into matrix form

Q1. $2x^2 + 3y^2 + 4z^2 + 8xy + 4yz - 2zx$

Soln

$$A = \begin{bmatrix} 2 & 4 & -1 \\ 4 & 3 & 2 \\ -1 & 2 & 4 \end{bmatrix}$$

Q2.

$$A = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 5 \end{bmatrix}$$

Write the quadratic form of this matrix

Soln

$$-x^2 + 3y^2 + 5z^2 + 4xy + 2xz + 4yz$$

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Q. Diagonalise the matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{bmatrix}$$

Also find A^3

Soln Characteristic eqn $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & -1 & 2 \\ 0 & 2-\lambda & -1 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$-\lambda^3 + 6\lambda^2 - (6+3+2)\lambda + 6 = 0$$

$$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$(3-\lambda)(1-\lambda)(2-\lambda) = 0$$

$$\boxed{\lambda = 1, 2, 3} \rightarrow \text{Eigen values}$$

Eigen vector

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 1-\lambda & -1 & 2 \\ 0 & 2-\lambda & -1 \\ 0 & 0 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

For $\lambda = 1$

$$0x_1 - y_1 + 2z_1 = 0$$
$$0x_1 + y_1 - z_1 = 0$$
$$0x_1 + 0y_1 + 2z_1 = 0$$

$$\frac{x_1}{1-2} = \frac{y_1}{0} = \frac{z_1}{0}$$

$$(-1, 0, 0)$$

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For $\lambda = 2$

$$\begin{bmatrix} 1-\lambda & -1 & 2 \\ 0 & 2-\lambda & -1 \\ 0 & 0 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = 0$$

$$-x_2 - y_2 + 2z_2 = 0$$
$$0x_2 + 0y_2 - z_2 = 0$$
$$0x_2 + 0y_2 + z_2 = 0$$

$$z_2 = 0$$

$$-x_2 = y_2$$
$$\frac{x_2}{1} = \frac{y_2}{-1}$$
$$(1, -1, 0)$$

For $\lambda = 3$

$$-2x_3 - y_3 + 2z_3 = 0$$
$$0x_3 - y_3 - z_3 = 0$$

$$\frac{x_3}{1+2} = \frac{y_3}{-2} = \frac{z_3}{2}$$
$$(3, -2, 2)$$

$$B = \begin{bmatrix} -1 & 1 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$B : I = \begin{bmatrix} -1 & 1 & 3 & : & 1 & 0 & 0 \\ 0 & -1 & -2 & : & 0 & 1 & 0 \\ 0 & 0 & 2 & : & 0 & 0 & 1 \end{bmatrix}$$

$-R_1$

$$\sim \left[\begin{array}{ccc:ccc} 1 & -1 & -3 & -1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array}\right]$$

$-R_2$

$$\sim \left[\begin{array}{ccc:ccc} 1 & -1 & -3 & -1 & 0 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_1 + R_2$

$$\sim \left[\begin{array}{ccc:ccc} 1 & 0 & -1 & -1 & -1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array}\right]$$

$1/2\ R_3$

$$\sim \left[\begin{array}{ccc:ccc} 1 & 0 & -1 & -1 & -1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/2 \end{array}\right]$$

$R_1 + R_3 \quad R_2 - 2R_3$

$$\sim \left[\begin{array}{ccc:ccc} 1 & 0 & 0 & -1 & -1 & 1/2 \\ 0 & 1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1/2 \end{array}\right]$$

$$B^{-1} = \begin{bmatrix} -1 & -1 & 1/2 \\ 0 & -1 & -1 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$B^{-1}AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$A^n = BD^nB^{-1}$$

$$A^3 = \begin{bmatrix} -1 & 1 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 27 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1/2 \\ 0 & -1 & -1 \\ 0 & 0 & 1/2 \end{bmatrix}$$

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Q. Find the characteristic eqn of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

Q Find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$